Neha Navarkar

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Summary

Robotics engineer specializing in SLAM, robot learning, and embedded systems. Experienced in building full autonomy stacks for mobile manipulators and sim-to-real locomotion systems, with a track record of deploying research-grade hardware solutions and leading cross-functional engineering projects.

Education

New York University — New York, NY
M.S. in Robotics and Mechatronics Engineering — GPA: 3.7/4.0 — *2024 – 2026*

D Y Patil College of Engineering, Akurdi — Pune, India
B.E. in Electronics Engineering — GPA: 9.18/10 — *2019 – 2023*

Skills

Autonomy & Robotics: ROS2, ZED 2i, ZED SDK (VIO/SLAM), A*/D*-Lite, voxel mapping, trajectory planning, sensor fusion
Robot Learning: Isaac Lab, PPO/RL, sim-to-real transfer, PyTorch, JAX, CUDA
Perception: OpenCV, YOLO, point-cloud processing, Rerun, Viser
Embedded & Hardware: FOC motor control, ESP32, Jetson, Qi wireless, UART/I2C/SPI, Fusion 360
Dev Tools: Python, C++, C, Docker, Linux, Git, Gazebo

Experience

Research Assistant — CILVR Lab (Generalizable Robotics & AI) — New York, NY
New York University — *May 2025 – Present*

- Built a full autonomous navigation stack for YOR using ZED 2i and Jetson/NUC: ZED SDK visual-inertial odometry, voxel mapping at 5 Hz, weighted A* planning, and pure-pursuit control.
- Achieved 12 mm odometry accuracy over 10 loops, <1 s obstacle replanning, and 16 mm end-effector deviation during whole-body motion.
- Contributing author, *YOR: Your Own Mobile Manipulator* (arXiv:2602.11150) — open-source $<10K mobile manipulator platform for whole-body coordination and autonomous navigation research.

Graduate Assistant — VIP STEP & MakerSpace — New York, NY
New York University — *Jan 2025 – Present*

- Designed gait patterns and embedded perception modules for bipedal locomotion on uneven terrain; authored SOPs and led hands-on training for 50+ students on UR5 robotic arm operation.

Graduate Apprentice — Chassis Systems — Chakan, India
Bosch — *Sep 2023 – Aug 2024*

- Won 1st place at Bosch's MDP Global Championship for a Python/MATLAB sensor-fusion analytics pipeline that reduced measurement discrepancies by 20%; led end-to-end delivery of 5 digitization projects using SQL and Power BI.

Projects

Go2 RL Locomotion | *Isaac Lab, PPO/RSL-RL, Python, Unitree Go2*

- Trained Unitree Go2 in Isaac Lab (NYU Greene HPC) with torque-based PD control, gait-phase clock inputs, multi-term reward shaping, and per-episode friction randomization ($\mu_v \sim \mathcal{U}(0, 0.3)$) for sim-to-real robustness.
- Extended the project by implementing an uneven terrain curriculum to further enhance locomotion robustness across varied surface conditions.

CPR Training Machine | *Raspberry Pi 5, AK60-6 BLDC, FOC Motor Control, Python/C*

- Portable AHA-compliant CPR device: slider-crank mechanism with FOC torque feedback achieves compression depth within 5% of target. Stack: C/Python, ZeroMQ IPC, Kivy GUI, OpenCV patient placement, breathing detection via current sensing.

Awards & Hackathons

Best Agentic Use — Innate Robotics Le Robot Hackathon 2025

- Developed an OpenAI-powered robotic poker player with gesture recognition for bet execution and real-time win-rate strategy adaptation.